

EMIR BEKAR

Mechatronics Engineer · Game Developer · Content Creator

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SUMMARY

Mechatronics Engineering graduate (100% English) with strong mathematical and analytical foundations, award-winning senior design research, and two IEEE-published papers in multi-agent robotics and AI. Hands-on experience developing mobile 3D/2D games in Unity, creating custom 2D pixel art and 3D assets, and managing an active 12,000+ subscriber YouTube community with direct insight into player retention and content iteration. Passionate about mobile casual games and eager to drive engaging product features and analytics at Peak.

EDUCATION

Istanbul Bilgi University

Jul 2026

Bachelor of Science in Mechatronics Engineering (100% English)

Istanbul, Turkey

- 🏆 **Best Senior Design Project Award:** "Adaptive Formation Control for Collaborative Swarm Robots" (Presented at HORA Congress 2026, published on IEEE Xplore).

FEATURED PROJECTS

Match 3D Mobile Game

Unity 6 · C# · Blender · URP

- Developed a hybrid-casual 3D matching mobile game with physics-based matching and zero-GC object pooling for stable 60 FPS.
- Designed core retention and monetization meta-game loops: Daily Rewards, Lucky Spin wheel, and in-game economy store.
- Modeled and textured modular 3D assets in Blender for mobile visual clarity. [[GitHub](#) | [Video Demo ↗](#)]

2D Pixel Art Indie Game (In Development)

Unity 2D · C# · LibreSprite · Pixel Art

- Hand-crafting original 2D pixel art character sprites, frame-by-frame animations, environment tilesets, and modular visual effects.
- Programming responsive 2D physics controllers, state machines, and interactive world gameplay mechanics from scratch.

AC-130 Gunship Simulation (In Development)

Unity 6 · C# · Custom Shaders

- Close air support tactical simulation featuring custom FLIR thermal vision shaders, dynamic terrain destruction, and avionics HUD telemetry. [[GitHub](#) | [Video Demo ↗](#)]

PUBLICATIONS (IEEE XPLORE · HORA CONGRESS 2026)

- Adaptive Formation Control for Collaborative Swarm Robots (IEEE 2026):** Leader-follower formation switching via FSM and fuzzy-logic ACC control eliminating stop-and-go oscillations, validated in Webots simulation. [[IEEE Xplore #11537161 ↗](#)]
- Multi Agent Reinforcement Learning Based Swarm Navigation (IEEE 2026):** Autonomous swarm coordination and collision-free path optimization using Multi-Agent Reinforcement Learning (MARL). [[IEEE Xplore #11537199 ↗](#)]

COMMUNITY & CONTENT CREATION

Emiukob – YouTube Creator & Mod Developer

12,000+ Subscribers · youtube.com/@emiukob

- Built a gaming and tech community to 12,000+ subscribers; analyzed audience retention curves, drop-off rates, and CTR to iteratively optimize content pacing.
- Developed and published "Hologram Projector" 3D hologram Minecraft mod in Java on CurseForge. [[CurseForge ↗](#)]

SKILLS & CERTIFICATIONS

Game & Product: Mobile casual / hybrid-casual games, 3D match mechanics, level pacing, FTUE, player retention & meta-game loops

Art & Technical Tools: Unity 6 / 2D (C#), 2D Pixel Art & Animation, LibreSprite, Blender (3D), Python, Java, JavaScript, Git/GitHub

Analytical: Data interpretation, A/B testing mindset, mathematical modeling, fuzzy logic, reinforcement learning, Webots

Certifications: Basic Python for Business Analytics (Arizona State University) | UAV 0/1 Pilot License (SHGM)